Assignment JC4

Handong Honor Code

- It is your responsibility to understand and adhere to the Handong Honor Code.
- Unauthorized copying of coding assignments, homework, or creative work will result in a failing grade (F) for both the individual who copies and the individual who shares the work.

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IMPORTANT NOTES

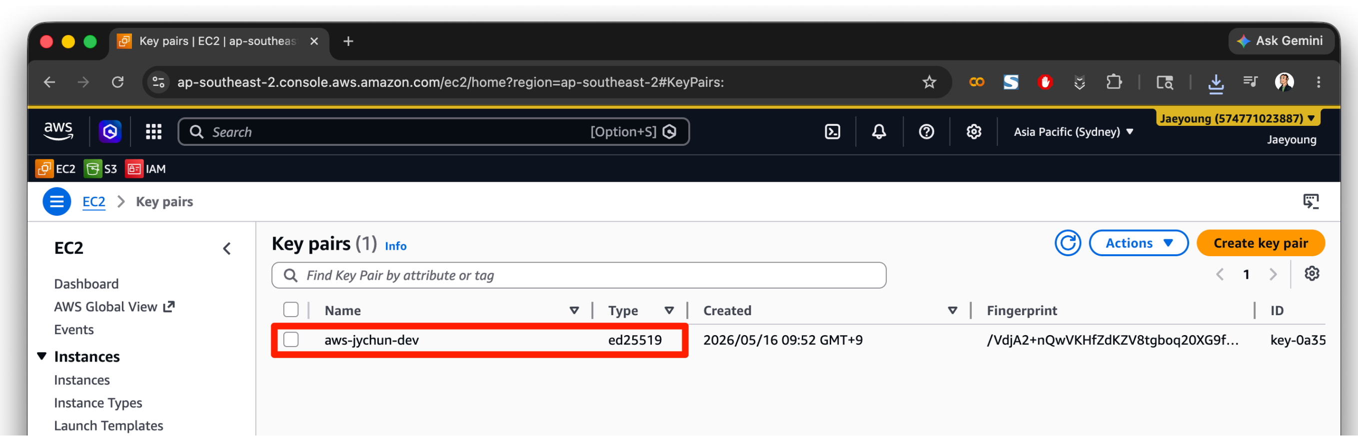
Ensure that you assign the `Owner` tag to all the cloud resources you create, with the value set to your *student ID (학번)*.

You will be asked to take a screenshot for each task. When taking the screenshots, there are a couple of important points to keep in mind:

1. Ensure to include the entire screen in the capture, not just the relevant section. This is to ensure that we can verify the context of the screenshot and confirm that it was taken by you.
2. You will be provided with an example of how the screenshot should look like for each task. Please follow the format shown in the examples when taking your screenshots, especially if there is a *red box* or *red arrow* highlighting a specific area. The red box or arrow is meant to indicate the specific section that we want you to focus on, but the entire screen should be visible in the screenshot.
3. Ensure that your username and account number are visible on the top right corner of the AWS Management Console in all screenshots. This is to verify that the screenshots were taken from your account and to ensure that you are the one performing the tasks.

1. Creating Key Pair

Create a key pair in AWS EC2 and take a screenshot of the key pair details. Save the screenshot as `학번-key-pair.png`.

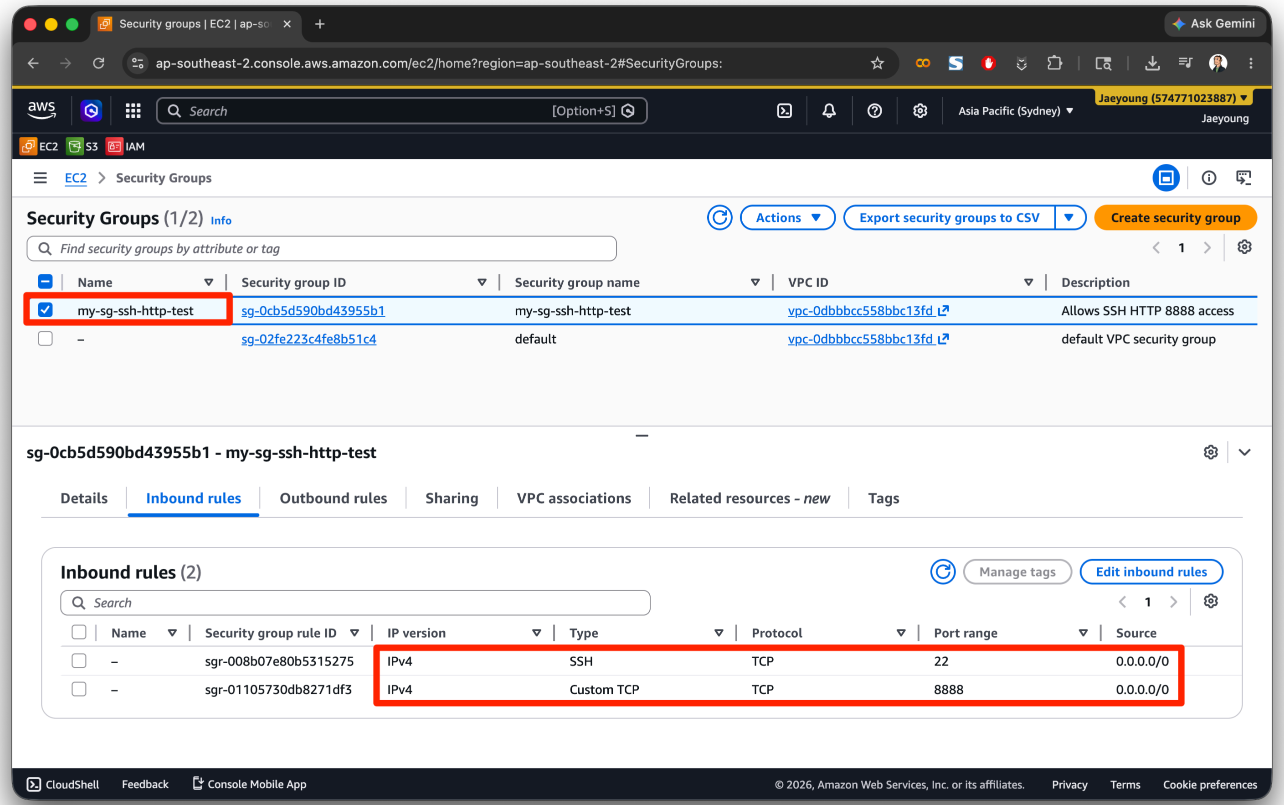


2. Creating Security Group

Create a security group in AWS EC2 with the following rules:

- Inbound Rules:
    - TCP 22 (SSH), Source: anywhere IPv4
    - HTTP 8888 (JupyterLab), Source: anywhere IPv4

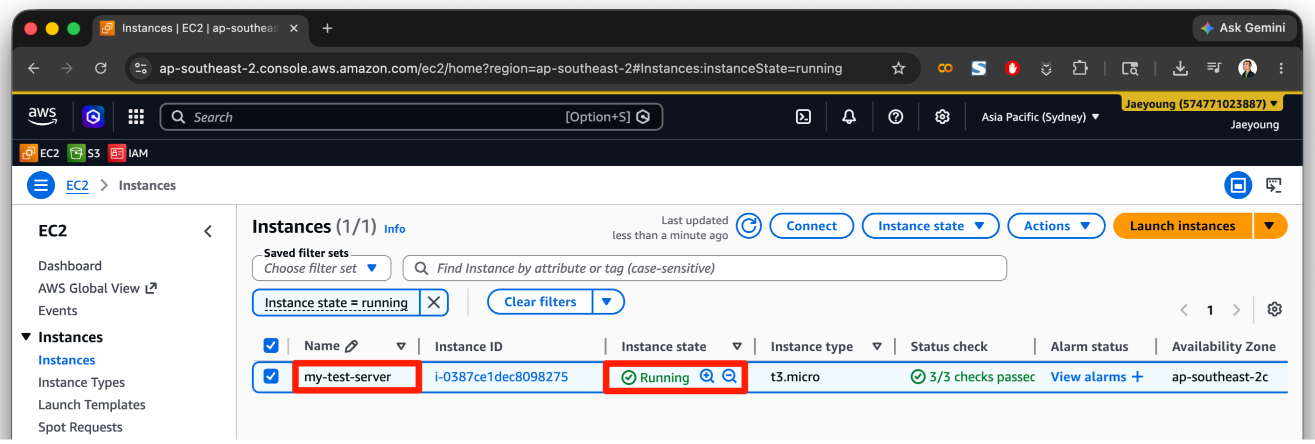
Take a screenshot and save it as `security-group.png`.



3. Launching EC2 Instance

Launch an EC2 instance using the key pair and security group you created in the previous steps.

Take a screenshot once the instance is running and save it as `ec2-running.png`.



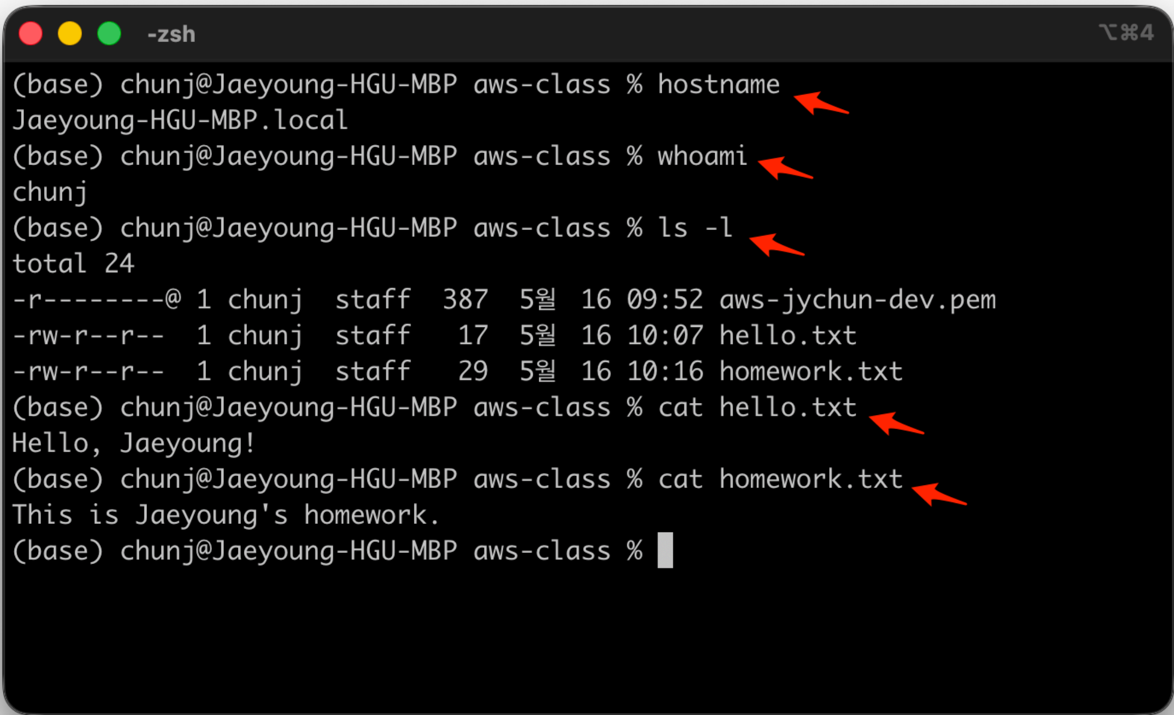
4. Secure File Transfer

1. Create a `hello.txt` file on your local machine with the content `Hello, OOO` (`OOO` should be replaced with your name).
2. Use `scp` to securely transfer the `hello.txt` file to your EC2 instance.
3. Create a `homework.txt` file on your EC2 instance with the content `This is OOO's homework"` `(OOO` should be replaced with your name).
4. Use `scp` to securely transfer the `homework.txt` file from your EC2 instance back to your local machine.

4-1. Screenshot of Your Local Terminal

Run the commands shown below from *local machine*, take a screenshot of the terminal, save it as `local.png`.

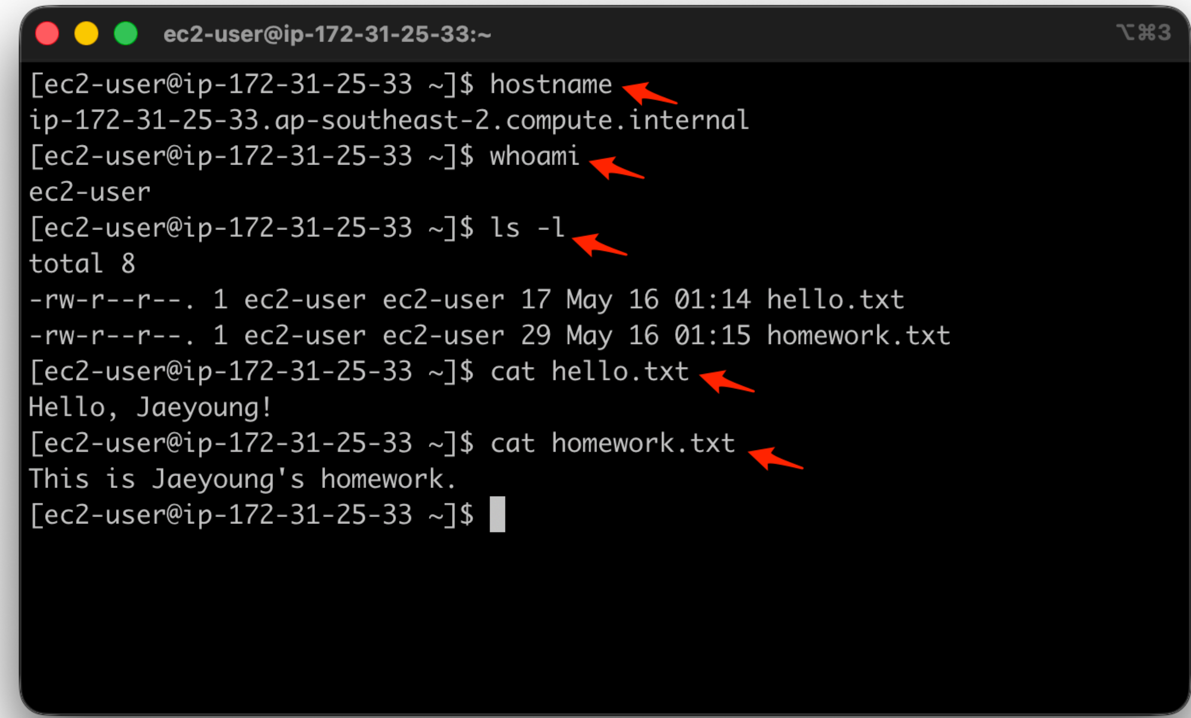
```
hostname
whoami
ls -l
cat hello.txt
cat homework.txt
```



4-2. Screenshot of Your EC2 Terminal

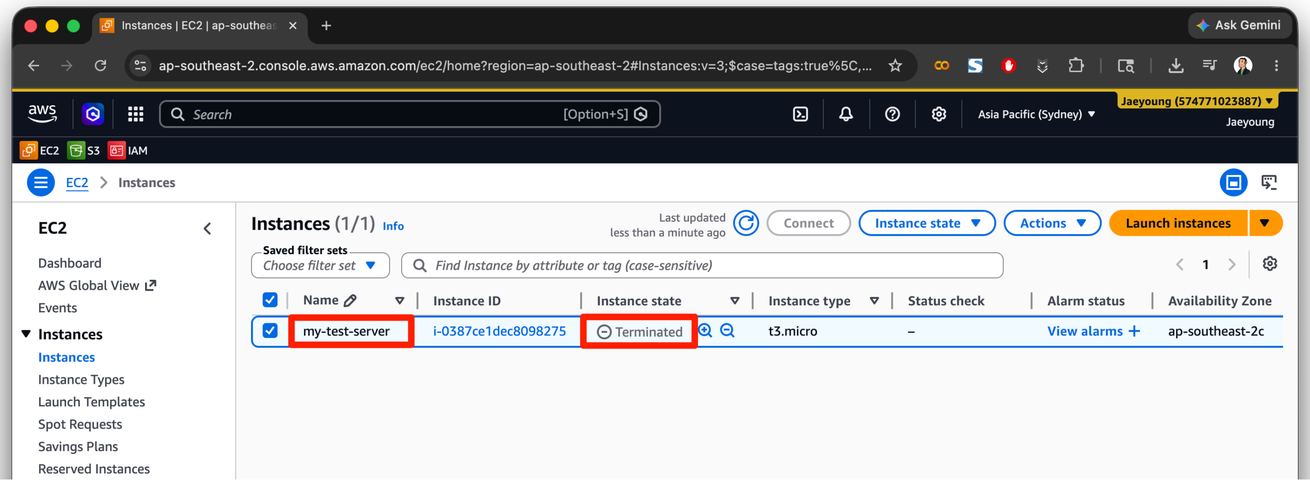
Run the commands shown below from *your EC2 instance*, take a screenshot of the terminal, save it as `remote.png`.

```
hostname
whoami
ls -l
cat hello.txt
cat homework.txt
```



5. Terminating EC2 Instance

After completing the above tasks, please terminate your EC2 instance to avoid unnecessary costs. Take a screenshot of the EC2 dashboard showing that your instance has been terminated, and save it as `ec2-terminated.png`.



Submission

When submitting, please package all the files into a single ZIP archive named `학번.zip` and submit it.

Example of submission format:

```
22316316.zip
├── key-pair.png
├── security-group.png
├── ec2-running.png
├── local.png
├── remote.png
└── ec2-terminated.png
```